Lecture 23c: GANs in practice

Control, translation, and how to score a generator

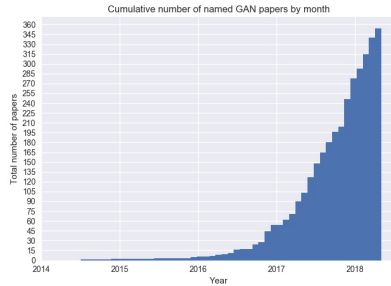
Parts of this chapter adapt LMU I2DL (slds-lmu), CC BY 4.0.

From "something" to "that"

L23b left us with a generator that draws *a* digit. Useful as a demonstration, useless as a tool - you cannot ask it for anything.

Everything GANs became famous for is about **control**: give me a seven, turn this sketch into a photograph, make this face older.

And one problem we have been dodging since L23: with no likelihood to report, **how does anyone claim one generator is better than another?**



LMU I2DL, CC BY 4.0

A few years of the literature. We take four ideas from it, not forty.

Outline

Asking for something specific

Translating images

Scaling up

Scoring a generator

Asking for something specific

One extra input, in both networks.

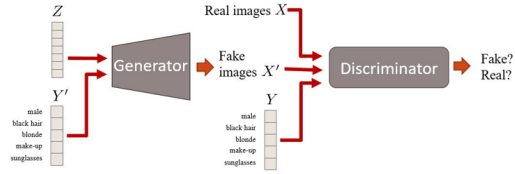
Conditional GANs

Feed the label **y** to the generator alongside **z**, so it knows what to draw. That alone is not enough.

The label must go into D as well. Otherwise D only asks "is this a real digit?", G can answer with any digit, and the label is free to ignore.

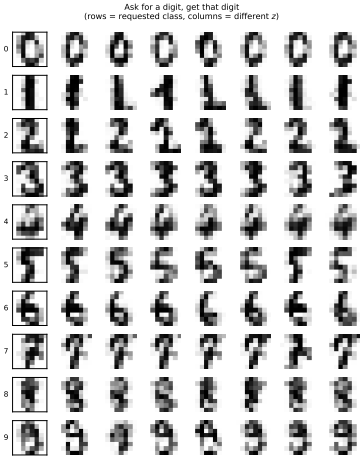
Conditioning D changes the question to "is this a real *seven*?" - and now ignoring the label costs G something.

Mirza & Osindero (2014). Compare with L30: diffusion had the same problem and needed guidance on top. A GAN gets obedience from the critic instead.



LMU I2DL, CC BY 4.0

Our own conditional GAN



Same 8×8 digits, same architecture as L23b, with the label added to both networks. Rows are the **requested** class; columns are different **z**.

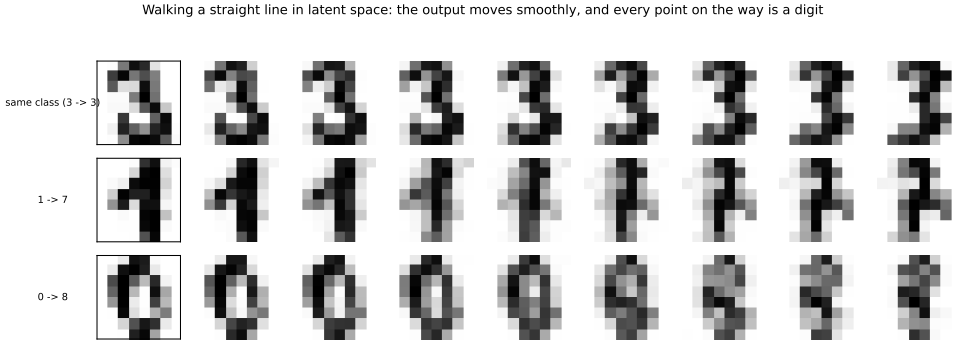
It obeys. And within each row the samples genuinely differ - conditioning bought control *without* collapsing variety.

Look at row 8. Mushier and less consistent than the rest - the same class the unconditional model under-covered in L23b.

Two independent runs agreeing on which digit is hardest is not a coincidence; 8 is the most structurally complex glyph at this resolution.

The latent space is smooth

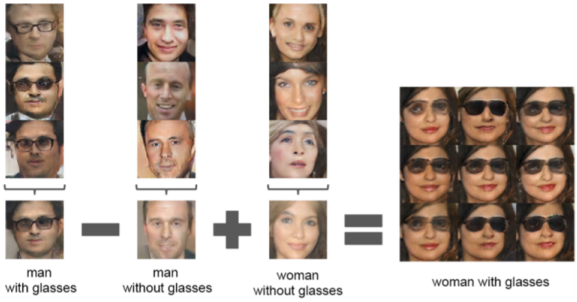
Walk in a straight line from one latent code to another and decode every point on the way:



Nothing forced this. We never trained on intermediate points. Yet the path between two digits passes through plausible digits rather than noise.

This is what makes latent-space **editing** possible at all, and it is the property StyleGAN later industrialized. **Directions in z mean something.**

Directions carry meaning



LMU I2DL, CC BY 4.0

Average the latent codes of several "man with glasses", subtract "man without glasses", add "woman without glasses" - and decode. You get a woman with glasses.

You have seen this structure before - in [54], where $king - man + woman \approx queen$ in a word-embedding space trained on nothing but next-word prediction.

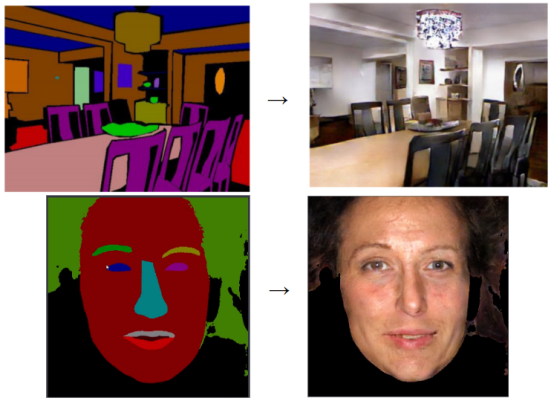
And you will meet it again in L30, where an image-text model learns a space in which (hat) − (no hat) picks out the word "hat".

Three unrelated training objectives, all producing spaces where *directions are concepts*. Nobody designed that in.

Translating images

If the input is an image instead of noise, the same machinery does something else.

Paired translation



Replace G's noise input with an **image**: a segmentation map, a sketch, a daytime photo. D sees the *pair* and judges whether the output belongs with that input.

pix2pix (Isola et al., 2017). Add a reconstruction term so the output stays faithful, and let the adversarial term supply the realism the reconstruction term would have blurred away.

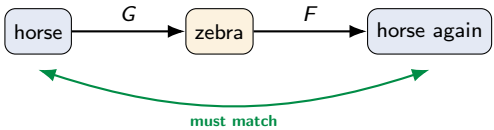
That split is worth noticing: the L1 term handles *correctness*, the critic handles *plausibility*. It is a direct answer to the VAE's blur.

LMU I2DL, CC BY 4.0

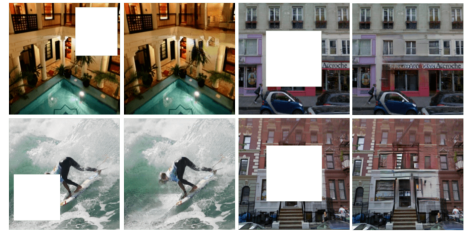
Unpaired translation, and the trick that made it work

pix2pix needs matched pairs of the same scene. For horses and zebras, no such dataset exists.

CycleGAN (Zhu et al., 2017) trains two generators, $G : X \to Y$ and $F : Y \to X$, and adds one constraint:



Cycle consistency: translate there and back, and you must land where you started. Without it, G could map every horse to the same zebra and still fool D - mode collapse with extra steps.



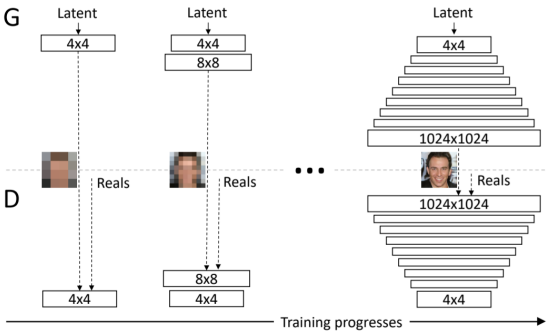
Inpainting. The white squares are holes cut out of the photo - the input. LMU I2DL, CC BY 4.0

The same image-in, image-out setup covers **inpainting** (fill the hole), super-resolution and colourization. Adversarial losses are still standard in super-resolution today.

Scaling up

How GANs got from blurry 32x32 to photorealistic faces.

Grow the problem gradually



LMU I2DL, CC BY 4.0

Training a GAN directly at 1024^2 fails - D finds trivial tells and G never catches up.

Progressive growing (Karras et al., 2017): start both networks at 4×4, train to convergence, then fade in another layer at double the resolution. Repeat.

Each stage is an easy problem given the one before it. A curriculum, applied to a training procedure that cannot handle the full task cold.

StyleGAN, and the peak of the GAN era



StyleGAN (Karras et al., 2019). None of these people exist.

StyleGAN restructured the generator: instead of feeding **z** in at the bottom, it maps **z** to an intermediate space **w** and injects it at *every* resolution.

Coarse layers control pose and face shape; fine layers control hair texture and freckles. So you can change one without disturbing the other.

This is when GANs were unambiguously state of the art for images - roughly 2019 to 2021. You already saw these faces in L19, unexplained. Now you know what produced them.

And you know why they were hard to train, which is why what came next displaced them.

Scoring a generator

No likelihood, no validation loss. Now what?

Why the usual tools are unavailable

A GAN never defines $p_G(\mathbf{x})$ explicitly. You can *sample* from it and nothing else - it is an **implicit** model.

So: no held-out likelihood, no perplexity, and as we saw, the training loss is not a quality signal either.

We need something computed from **samples alone**, and it has to punish both failure modes:

- outputs that do not look real
- outputs that look real but are all the same

A metric blind to the second would have scored our step-500 collapse as excellent.

Fréchet Inception Distance (FID) - the standard answer. Push real and generated images through a pretrained vision network, fit a Gaussian to each set of features, and measure the distance between those two Gaussians. Lower is better.

What FID does and does not tell you

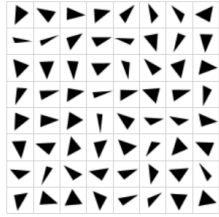
FID collapses two different failures into one number. A model with poor realism and a model with poor diversity can score the same.

All four panels generate triangles. **(b)** draws them well but in one orientation - realistic, no coverage. **(c)** covers every orientation but draws them badly.

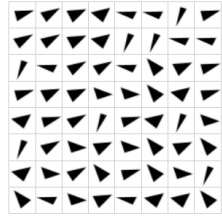
Hence **precision and recall for generative models** (Sajjadi et al., 2018): **precision** = are the samples realistic, **recall** = does the model cover the data. Two axes, separately reportable.

Our digit GAN: decent precision, slightly weak recall - the missing 8s.

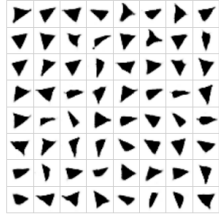
FID is a comparator, not a constant: it inherits its feature network's biases.



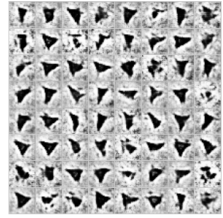
(a) High precision, high recall



(b) High precision, low recall



(c) Low precision, high recall



(d) Low precision, low recall

Where this leaves GANs

Still winning: anything that needs **one forward pass**. Real-time super-resolution, upscaling in games, on-device photo enhancement, audio synthesis.

A GAN generates in a single pass. Diffusion needs tens.

Displaced for text-to-image, and the reason is the whole of L23b: an unstable objective that resists scaling.

Diffusion offered a **stationary loss** that reliably improves with compute - exactly what a lab with a large budget wants.

The lesson is not "GANs lost". It is that a method's ceiling is often set by how reliably it trains, not by how good its best result is. GANs produced photorealistic faces before diffusion did - and lost anyway, on trainability.

Recap

1. Conditioning needs the label in **both** networks, or the generator ignores it.
2. Our conditional GAN obeys, keeps variety, and is weakest on the same digit the unconditional one under-covered.
3. Latent space is smooth and its directions carry meaning - the third time this course has met that structure.
4. Image-in instead of noise-in gives translation; cycle consistency makes it work without paired data.
5. Progressive growing and StyleGAN took GANs to photorealism.
6. With no likelihood, quality is measured from samples: FID, and better, precision/recall as separate axes.

Next: you now have two of the three generative families - the VAE's blur and the GAN's instability.

Both are solved by a third approach that trains as simply as the VAE and generates as sharply as a GAN. That is the diffusion chapter (L27-L31).

Before it, the attention chapter (L24) picks up the other thread from [54].